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Options and Futures

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Hedging in Black-Scholes

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- The Greeks
- Delta-hedging in discrete time
- Hedging a book in practice

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The Greeks

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- The BS pricing formulas for European calls and puts are rather complicated functions of many variables and parameters (the variables are S and t , and the parameters are σ, r, T and K).
- Denoting the derivative price by V, we now look at partial derivatives of $V(S, t, \sigma, r, T, K)$, also known as the "Greeks". Each Greek gives us the sensitivity of the derivative price to one argument, holding the other arguments constant.
- We focus on the most well-known Greeks. There are many more cross-partial derivatives that are relevant for a trader.
  - For instance, how does the effect of volatility depend on time-to-expiration?

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Delta

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- Delta tells us how much the option value changes for a small change in the price of the underlying, other things being equal:

$$\delta = \frac{\partial V}{\partial S}.$$

- δ is also a hedge-ratio. This can be seen from the delta-hedging argument used in the derivation of the BS PDE.
- In the BS model, the delta of a European call can be read off from the BS formula:

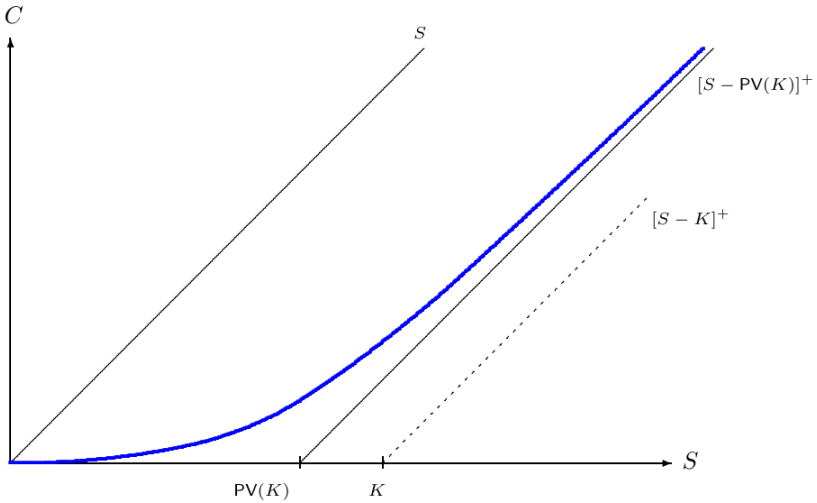
$$\delta = \Phi(d_1).$$

  To hedge a short position of 1 in the call, buy $\Phi(d_1)$ units of the underlying.
- This is called a delta-neutral position, with a position delta of 0.

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Delta is the slope of BS price

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Properties of delta

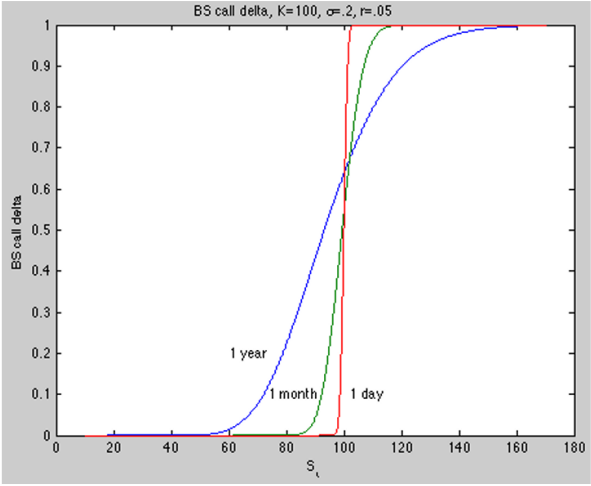
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- Recall that
$$d_1 = \frac{\log(S/K) + (r + \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}.$$
- Call delta for deep ITM and OTM options:
$$\lim_{S\to\infty} \delta = 1 (d_1 \to \infty),$$
$$\lim_{S\to 0} \delta = 0 (d_1 \to -\infty).$$
- Call delta at maturity:
$$\lim_{t\to T} \delta = \begin{cases} 1 & \text{if } S > K \quad (d_1 \to \infty), \\ 1/2 & \text{if } S = K \quad (d_1 \to 0), \\ 0 & \text{if } S < K \quad (d_1 \to -\infty). \end{cases}$$
- By put-call parity, $\delta_{\text{call}} = \delta_{\text{put}} + 1$
- $0 < \delta_{\text{call}} < 1, -1 < \delta_{\text{put}} < 0.$

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Call delta for different times to maturity

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Gamma

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- The gamma of a derivative measures by how much its delta changes for a small change in the price of the underlying, other things being equal:

$$\Gamma = \frac{\partial \delta}{\partial S} = \frac{\partial^2 V}{\partial S^2}.$$

- In the BS model, the gamma of a European call is

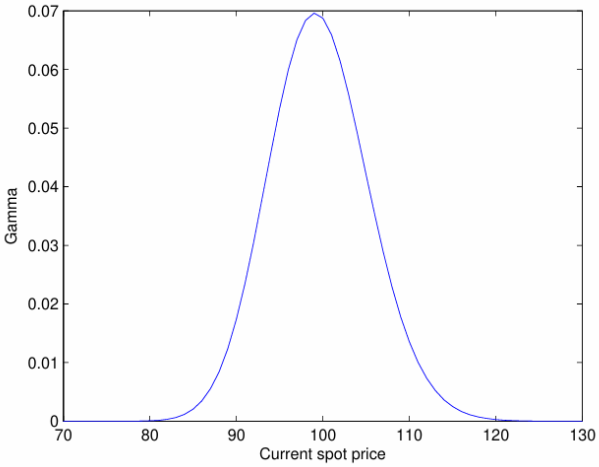
$$\Gamma = \frac{\partial \delta}{\partial S} = \Phi'(d_1)\frac{\partial d_1}{\partial S} = \frac{1}{S\sigma\sqrt{T-t}}\phi(d_1) > 0,$$

where ϕ is the standard normal density, $\phi(x) = \Phi'(x)$

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Gamma as a function of S

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Assumptions: $\sigma = 0.2$, $T = 1$, $K = 100$, $r = 0.05$

Properties of gamma

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- Gamma is a positive, bell-shaped function of S , strictly increasing in S up to a maximum and then strictly decreasing
  -It is maximized approximately ATM.
- Gamma for deep ITM and OTM European call options:

$$\lim_{S\to\infty} \Gamma = \lim_{S\to 0} \Gamma = 0.$$

- Gamma at expiration:

$$\lim_{t\to T} \Gamma = \begin{cases} 0 & \text{if} \quad S > K, \\ \infty & \text{if} \quad S = K, \\ 0 & \text{if} \quad S < K. \end{cases}$$

- $\Gamma_{\text{call}} = \Gamma_{\text{put}}$ by put-call parity.

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Feedback effect of delta-hedging on volatility

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- As the price of the underlying moves, the delta of a position changes. In order to remain delta-neutral, it is necessary to revise the position over time.
- Gamma is positive for both calls and puts: Delta increases with the price of the underlying.
- So delta-hedging a short position in a call or a put requires buying the underlying when its price goes up (increasing the long position in the case of a call, and reducing the short position in the case of a put), and selling the underlying after a price drop.
- Hence delta-hedging by institutions who write options or trading by institutions who try to replicate call/put options payoffs (with dynamic replicating strategies) tend to increase market volatility.
  - e.g. portfolio insurance (which involves synthetically replicating a put) was thought to have accelerated the fall in stock prices in the crash of 1987.

Vega

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- The impact of volatility σ is captured by vega:

$$\nu := \frac{\partial V}{\partial \sigma}.$$

- In the BS model, we have

$$\mathcal{V} = S\phi(d_1)\sqrt{T-t}$$

  for both calls and puts, by put-call parity.

- The option price is increasing in σ
  - Intuition for calls: Upward price movements are larger and yield higher payoff; downward price movements are also larger, but they do not matter because the payoff is bounded below at zero.
- The volatility of the underlying is assumed to be constant in BS, so vega should be understood as a comparative static.

Properties of vega

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- Vega is a lot like gamma. Both are positive, highest ATM (approximately), and very small for deep ITM or OTM options.
- Note that both calls and puts converge to their intrinsic value as volatility goes to zero:

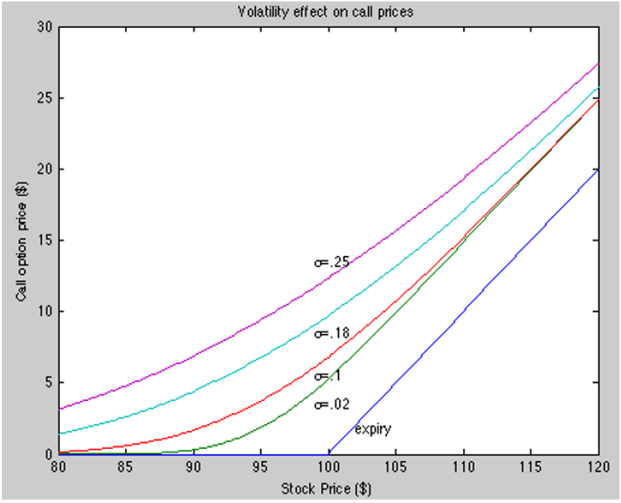
$$\lim_{\sigma \to 0} C = \left[S - e^{-r(T-t)} K \right]^+,$$
$$\lim_{\sigma \to 0} P = \left[e^{-r(T-t)} K - S \right]^+.$$

- Interpretation: options derive their time value from volatility.

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Call price for different values σ

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- Even if the stock price remains unchanged, as the maturity date approaches, the mere passing of time creates profits or losses in option positions.
- We define an option's theta as

$$\Theta := \frac{\partial V}{\partial t}.$$

- In the B-S model, we have

$$\Theta_{\text{call}} = -re^{-r(T-t)}K\Phi(d_2) - \frac{S\sigma}{2\sqrt{T-t}}\phi(d_1) < 0,$$
$$\Theta_{\text{put}} = re^{-r(T-t)}K\Phi(-d_2) - \frac{S\sigma}{2\sqrt{T-t}}\phi(d_1).$$

Properties of theta

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- The first term, in which $T - t$ appears attached to r , is the discount effect. The shorter the time to maturity, the less K is discounted. Hence, the call is worth less and the put is worth more.
- The second term, in which $T - t$ appears attached to σ, is the volatility effect. The shorter the time to maturity, the lower is the variance of S_T given S_t , and hence the lower is the upside potential. This implies a lower value of both the call and the put.
- For calls, theta is negative and highest in absolute value ATM (approximately).
- For puts, theta is typically negative but can occasionally be positive (for deep ITM puts).

Rho

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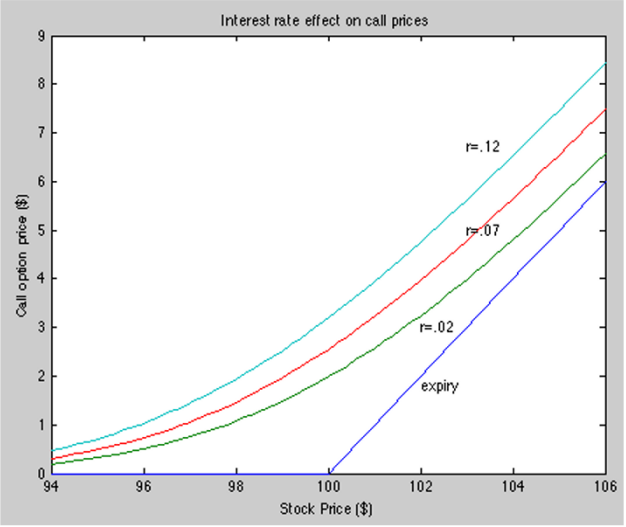
- Rho measures the sensitivity of the BS option price to a small change in the interest rate, assumed to stay constant until the option's maturity..
- The rho for a call and put in the BS model are

$$\rho_{\text{call}} = K(T-t)e^{-r(T-t)}\Phi(d_2) > 0,$$
$$\rho_{\text{put}} = -K(T-t)e^{-r(T-t)}\Phi(-d_2) < 0.$$

- Everything else equal, a higher interest rate means that the cost K of obtaining the stock at maturity is lower in present value terms, making the call more valuable.
- The opposite is true for a put: The present value of the receipt K is lower with a higher interest rate, meaning that the put effectively sells the underlying for less.

Call price for different values of r

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Signs of the Greeks for European calls and puts in the BS model

Greeks	Call	Put
$\delta = \partial V/\partial S$	positive	negative
$\Gamma = \partial^2 V/\partial S^2$	positive	positive
$\mathcal{V} = \partial V/\partial \sigma$	positive	positive
$\Theta = \partial V/\partial t$	negative	+/-
$\rho = \partial V/\partial r$	positive	negative

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Delta-hedging in discrete time

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- The replicating strategy in Black-Scholes requires continuous rebalancing.
- In practice, trading costs are prohibitive.
- Traders adjust their hedge periodically.

Background: Theta vs gamma in the BS PDE

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- Suppose a trader sells a European call option and engages in delta-hedging.
- The trader collects the premium C_t, buys $\delta_t = \Phi(d_{1t})$ units of the underlying, and invests the proceeds $\beta_t = C_t - \delta_t S_t$ in the MMA (note that $\beta_t < 0$).
- The net cost is zero, i.e. the total value (or cost) of the position is

$$v_t = -C_t + \delta_t S_t + \beta_t = 0.$$

- Hence,

$$\begin{aligned} dv_t &= -dC_t + \delta_t dS_t + d\beta_t \\ &= -dC_t + \delta_t dS_t + r\beta_t dt \\ &= -dC_t + \delta_t dS_t + r(C_t - \delta_t S_t)dt. \end{aligned}$$

Theta vs gamma in the BS PDE (cont'd)

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- Itô's lemma (together with the GBM assumption) implies

$$\begin{aligned} -dC_t &= -\frac{\partial C}{\partial t}dt - \frac{\partial C}{\partial S}dS_t - \frac{1}{2}\frac{\partial^2 C}{\partial S^2}(dS_t)^2 \\ &= \left(-\Theta_t - \frac{1}{2}\sigma^2 S_t^2 \Gamma_t\right) dt - \delta_t dS_t. \end{aligned}$$

- Hence

$$\begin{aligned} dv_t &= -dC_t + \delta_t dS_t + r(C_t - \delta_t S_t)dt \\ &= \left[-\Theta_t - \frac{1}{2}\sigma^2 S_t^2 \Gamma_t + r(C_t - \delta_t S_t)\right] dt. \end{aligned}$$

- But $v_t = 0 \Rightarrow dv_t = 0$ (by no-arbitrage), i.e.

$$\Theta_t + \frac{1}{2}\sigma^2 S_t^2 \Gamma_t - r(C_t - \delta_t S_t) = 0.$$

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Theta as the "mirror image" of gamma

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- From the previous slide:

$$\Theta_t + \frac{1}{2}\sigma^2 S_t^2 \Gamma_t - r(C_t - \delta_t S_t) = 0.$$

- This is the Black-Scholes PDE.
- Note that the drift of the trader's short call option position is $-\Theta_t - \frac{1}{2}\sigma^2 S_t^2 \Gamma_t$
- Since $\Theta_t < 0$ (time decay)and $\Gamma_t > 0$. Θ_t plays in his favor by increasing the drift, while Γ_t plays against him by reducing the drift.
- The gain due to theta compensates for the loss due to gamma.
- If $r = 0$, the two are exactly offsetting:

$$\Theta_t + \frac{1}{2}\sigma^2 S_t^2 \Gamma_t = 0.$$

Theta vs gamma over a discrete time interval

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- Once again, consider a trader who sells a call at time t delta-hedges, and invests $C_t - \delta_t S_t$ in the MMA.
- Now fix the portfolio over a discrete (but small) time interval Δt
- Assume that the time interval is short enough to ignore compounding of interest.
- Then the trader's P&L, Δv_t , is (approximately)[1]

$$\Delta v_t \simeq -\Delta C_t + \delta_t \Delta S_t + r(C_t - \delta_t S_t)\Delta t.$$

- A second-order Taylor approximation of ΔC_t is

$$\begin{aligned}\Delta C_t &\simeq \frac{\partial C}{\partial t}\Delta t + \frac{\partial C}{\partial S}\Delta S_t + \frac{1}{2}\frac{\partial^2 C}{\partial S^2}(\Delta S_t)^2 \\ &= \Theta_t \Delta t + \delta_t \Delta S_t + \frac{1}{2}\Gamma_t (\Delta S_t)^2.\end{aligned}$$

[1]Note that δ_t is the call delta at time t, given by $\frac{\partial C}{\partial S}(S_t, t)$, while Δ denotes a discrete change (Δt is a discrete time interval, $\Delta S_t = S_{t+\Delta t} - S_t$,etc.)

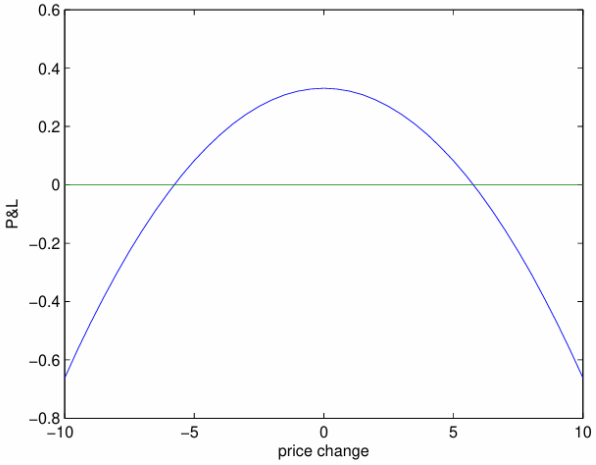
Theta vs gamma (cont'd)

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- Hence
$$\Delta v_t \simeq -\Theta_t \Delta t - \frac{1}{2}\Gamma_t (\Delta S_t)^2 + r(C_t - \delta_t S_t)\Delta t.$$
- Using the BS PDE,
$$-\Theta_t + r(C_t - \delta_t S_t) = \frac{1}{2}\sigma^2 S_t^2 \Gamma_t,$$
so that
$$\Delta v_t \simeq \frac{1}{2}\Gamma_t \left[(\sigma S_t \sqrt{\Delta t})^2 - (\Delta S_t)^2\right].$$
- Thus Δv_t is stochastic. It depends on ΔS_t
- Moreover, the impact of the realization of ΔS_t is scaled by Γ_t

Δv_t as a function of ΔS_t

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Assumptions: $S_t = 100$, $\sigma = 0.2$, $T - t = 1$, $K = 100$, $r = 0$, and $\Delta t = 1/12$.

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Discrete vs continuous hedging in the BS world

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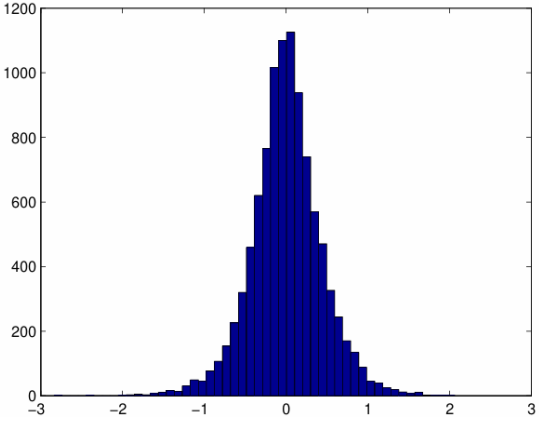
- If the delta-hedge is continuously adjusted over the lifetime of the option, dv_t is exactly zero, i.e. the P&L of a trader who delta-hedges continuously is zero.
- If all BS assumptions hold but the delta-hedge is adjusted discretely, the P&L is random.
- The realized P&L depends on the change in the stock price.
- The magnitude of the P&L is larger when Γ is high (i.e. when the option is close to the money).

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Distribution of P&L

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Distribution of trader's P&L based on 10,000 simulations



Assumptions: $\mu = 0.05$, $\sigma = 0.2$, $S_t = 100$, $r = 0.05$, $K = 100$ $T - t = 1$, $\Delta t = 1/250$

Hedging a book in practice

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- Suppose you have a book of positions on derivatives on the same underlying and you want to reduce the volatility of the value of your book as much as possible.
- How can you "immunize" the value of your portfolio against the sources of fluctuations in derivative prices?
- In the Black-Scholes model, the only source of fluctuations is movements in the price of the underlying
  - While we considered the case of a short position in a call in the preceding analysis, the same applies for any derivatives position. The BS PDE is valid for an arbitrary derivative payoff $g(S_T)$
- In practice, traders also hedge risks that are not at play in the Black-Scholes model, like volatility risk or interest rate risk.
- Moreover, traders typically do not hedge the entire book. Some part of it they may find too expensive to hedge, while another part they might want to leave unhedged for speculative reasons.

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Managing the risk of a book with the Greeks

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- The Greeks are useful tools for risk management.
- By linearity of differentiation, any Greek for a book is equal to the sum of the Greeks of all the positions in the book.
  - Suppose a book consists of a position n_i in security i. If the value of security i is V_i, the value of the book Π is

$$\Pi = \sum_i n_i V_i,$$

  - The sensitivity of the book with respect to a variable or parameter x is

$$\frac{\partial \Pi}{\partial x} = \sum_i n_i \frac{V_i}{\partial x}.$$

Gamma-hedging

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- As an approximation,

$$\Delta\Pi_t \simeq \Theta_t \Delta t + \delta_t \Delta S_t + \frac{1}{2}\Gamma_t (\Delta S_t)^2.$$

- Taking a delta-neutral position at discrete intervals of time leaves the value of the portfolio exposed to some risk.
- Since in practice it is not feasible to be continuously delta-hedged, a reinforcing strategy is to select positions that, in addition to being close to delta-neutral, have deltas that are relatively insensitive to movements in the price of the underlying, i.e. low gammas.
- A gamma-neutral position has a gamma of zero.

Gamma-hedging (cont'd)

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- If you want to be both delta- and gamma-neutral, you need to trade two assets whose values depend on the underlying, e.g. the underlying itself and a plain vanilla option (note that the gamma of the underlying itself is zero).
- Suppose your initial book has sensitivities δ^B and Γ^B , and denote δ^O and Γ^O the Greeks of a traded option. You can make your portfolio delta- and gamma-neutral by taking positions x and y in the underlying and in the option such that

$$\begin{aligned} \delta^B + x + y\delta^O &= 0, \\ \Gamma^B + y\Gamma^O &= 0. \end{aligned}$$

- Since δ^B and Γ^B vary over time, the hedge has to be adjusted to keep the portfolio delta- and gamma-neutral.

Vega-hedging

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- Though it is assumed to be constant in the BS model,the volatility of the underlying is likely to vary over time.
- As an approximation,

$$\Delta\Pi_t \simeq \Theta_t \Delta t + \delta_t \Delta S_t + \frac{1}{2}\Gamma_t (\Delta S_t)^2 + \mathcal{V}_t \Delta\sigma_t.$$

- For a given initial book, keeping a portfolio delta-, gamma-, and vega-neutral requires us to trade in three assets (e.g. the underlying and two derivatives).
- It is current practice to compute vega from the BS formula, even though BS rules out volatility risk by assumption. Though inconsistent, this practice works quite well: the BS vega is a good guide to monitoring exposure to volatility risk.

Remarks on hedging

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- Positions are never perfectly hedged in practice.
- Hedging is model dependent. For instance, the delta of a call is $\Phi(d_1)$ in BS but this is typically not true in another pricing model.
- Traders often engage in "out-of-model" hedging: Even though some parameter is taken as constant in the model, it can be considered as a source of risk, and the corresponding Greek can be used to monitor this risk.

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- Appendix A contains proofs of some results
- This material will not be examined

$\mathbb{E}[(X-K)^+]$ for X lognormal

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- To derive the Black-Scholes formula, we used the following result: Suppose X is lognormally distributed, and $\text{Var}(\log X) = \sigma^2(T-t)$. Then

$$\mathbb{E}[(X-K)^+] = \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2),$$

where Φ is the standard normal cdf and

$$d_1 : \frac{\log[\mathbb{E}(X)/K] + \sigma^2(T-t)/2}{\sigma\sqrt{T-t}}, \quad d_2 := d_1 - \sigma\sqrt{T-t}$$

- To simplify notation, let $\mathbb{E}(\log X) = \alpha$, and $\text{Var}(\log X) = \beta^2$ (i.e. $\beta := \sigma\sqrt{T-t}$)

Proof

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- We can write $\log X = \alpha + \beta z$, where z is standard normal. Thus $X = e^{\alpha+\beta z}$ and $\mathbb{E}(X) = e^{\alpha+\beta^2/2}$. Moreover, $X \geq K$ is equivalent to $\log X \geq \log K$,or

$$\begin{aligned} z &\geq \frac{\log K - \alpha}{\beta} \\ &= \frac{\log K - \log \mathbb{E}(X) + \beta^2/2}{\beta} \\ &= -\frac{\log[\mathbb{E}(X)/K] - \beta^2/2}{\beta} \\ &= -d_2. \end{aligned}$$

- Recall the standard normal density: $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$
- Let Φ be the standard normal cdf.

Proof(cont'd)

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Thus we have:

$$
\begin{aligned}
\mathbb{E}[(X-K)^+] &= \int_{-d_2}^{\infty} (e^{\alpha+\beta z} - K)\phi(z)dz \\
&= \int_{-d_2}^{\infty} e^{\alpha+\beta z}\phi(z)dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \int_{-d_2}^{\infty} \mathbb{E}(X)e^{-\frac{1}{2}\beta^2+\beta z}\cdot\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2-\beta}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_1}^{\infty}\phi(z-\beta)d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2).
\end{aligned}
$$

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Breeden-Litzenberger

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- Let $\varphi(\cdot)$ be the density of S_T under the risk-neutral probability Q The price of a call option at $t = 0$ as a function of the strike price K can be written as:

$$C(K) = e^{-rT}\mathbb{E}^Q[(S_T - K)^+] = e^{-rT}\int_K^\infty (x - K)\varphi(x)dx$$

- Result [Breeden-Litzenberger (1978)]:

$$\varphi(K) = e^{rT}C''(K).$$

Proof of Breeden-Litzenberger result

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Proof: We can write

$$e^{rT}C(K) = \int_K^\infty x\varphi(x)dx - K\int_K^\infty \varphi(x)dx.$$

Differentiating:

$$\begin{aligned} e^{rT}C'(K) &= \frac{d}{dK}\int_K^\infty x\varphi(x)dx - K\frac{d}{dK}\int_K^\infty \varphi(x)dx - \int_K^\infty \varphi \\ &= -K\varphi(K) - K[-\varphi(K)] - \int_K^\infty \varphi(x)dx \\ &= -\int_K^\infty \varphi(x)dx. \end{aligned}$$

Differentiating again:

$$e^{rT}C''(K) = -[-\varphi(K)] = \varphi(K).$$

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- Put-call parity (non-dividend paying stock):

$$C_t = P_t + S_t - e^{-r(T-t)}K$$

- Lognormal property of GBM:
  Suppose $dS_t = \mu S_t dt + \sigma S_t dW_t$. Then the conditional distribution of $\log S_T$ at time t is normal:

$$\log S_T \sim \mathcal{N}\left[\log S_t + \left(\mu - \sigma^2/2\right)(T-t), \sigma^2(T-t)\right]$$

- Black-Scholes formula:
  The price of a European call option at $t < T$ is given by

$$C(S,t) = S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2)$$

  where $\Phi(\cdot)$ is the standard normal cdf, and

$$d_1 = \frac{\log(S/K) + \left(r + \sigma^2/2\right)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}$$

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- Black-Scholes PDE:
  Consider a derivative on a stock with payoff $g(S_T)$ at time T, whose price at time $t \leq T$ is given by $F(S,t)$. The Black-Scholes PDE is

$$F_t(S,t) + rSF_S(S,t) + \frac{1}{2}\sigma^2 S^2 F_{SS}(S,t) - rF(S,t) = 0$$

- Standard normal density:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

- Expectation of the exponential of a normal random variable:
  If X normally distributed, then

$$\mathbb{E}[\exp(X)] = \exp\left[\mathbb{E}(X) + \frac{1}{2}\operatorname{Var}(X)\right]$$